

Template

Studentnames and studentnumbers here

2026-06-18

Set-up your environment

```
require(tidyverse)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.3      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```
require(yaml)
```

```
## Loading required package: yaml
```

```
require(eurostat)
```

```
## Loading required package: eurostat
```

```
require(maps)
```

```
## Loading required package: maps
```

```
##
```

```
## Attaching package: 'maps'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      map
```

```
require(patchwork)
```

```
## Loading required package: patchwork
```

Title Page

Include your names: Imane Ouahi, David Abelovic, Xi Hui, Rebecca Devasia Let's do it!!! Hi Rebecca

Include the tutorial group number: 5 2

Include your tutorial lecturer's name: Simon Loop

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

Include the following:

- Why is this relevant?
- ...

Part 2 - Data Sourcing

2.1 Load in the data

Preferably from a URL, but if not, make sure to download the data and store it in a shared location that you can load the data in from. Do not store the data in a folder you include in the Github repository!

```
gpg_data <- get_eurostat("tesem180")
```

```
## indexed 0B in 0s, 0B/sindexed 1.00TB in 0s, 4.44PB/s
```

```
## Table tesem180 cached at /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//RtmpuL83K5/eurostat/c7529
```

```
salary_data <- get_eurostat("nama_10_fte")
```

```
## indexed 0B in 0s, 0B/sindexed 1.00TB in 0s, 1.85PB/s
```

```
## Table nama_10_fte cached at /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//RtmpuL83K5/eurostat/a4
```

```
saveRDS(gpg_data, "data/gpg_data")
saveRDS(salary_data, "data/salary_data")
salary_data <- readRDS("data/salary_data")
gpg_data <- readRDS("data/gpg_data")
```

midwest is an example dataset included in the tidyverse package

2.2 Provide a short summary of the dataset(s)

```
summary(gpg_data)
```

```
##          freq          unit          nace_r2          geo
## Length   :584   Length   :584   Length   :584   Length   :584
## N.unique  : 1   N.unique  : 1   N.unique  : 1   N.unique  : 39
## N.blank   : 0   N.blank   : 0   N.blank   : 0   N.blank   : 0
## Min.nchar : 1   Min.nchar : 2   Min.nchar : 7   Min.nchar : 2
## Max.nchar : 1   Max.nchar : 2   Max.nchar : 7   Max.nchar : 9
##
##   TIME_PERIOD      values
## Min.   :2007-01-01   Min.   : -1.30
## 1st Qu.:2011-01-01   1st Qu.: 9.70
## Median :2016-01-01   Median :14.45
## Mean   :2015-08-06   Mean   :13.87
## 3rd Qu.:2020-01-01   3rd Qu.:17.60
## Max.   :2024-01-01   Max.   :30.90
```

```
summary(salary_data)
```

```
##          freq          unit          geo          TIME_PERIOD
## Length   :1560   Length   :1560   Length   :1560   Min.   :1995-01-01
## N.unique  : 1   N.unique  : 2   N.unique  : 28   1st Qu.:2004-01-01
## N.blank   : 0   N.blank   : 0   N.blank   : 0   Median :2011-01-01
## Min.nchar : 1   Min.nchar : 3   Min.nchar : 2   Mean   :2010-06-16
## Max.nchar : 1   Max.nchar : 3   Max.nchar : 9   3rd Qu.:2018-01-01
##                                         Max.   :2024-01-01
##
##   values
## Min.   : 964
## 1st Qu.:14585
## Median :25700
## Mean   :97584
## 3rd Qu.:38114
## Max.   :7297665
```

In this case we see 28 variables, but we miss some information on what units they are in. We also don't know anything about the year/moment in which this data has been captured.

```
inline_code = TRUE
```

These are things that are usually included in the metadata of the dataset. For your project, you need to provide us with the information from your metadata that we need to understand your dataset of choice.

2.3 Describe the type of variables included

Think of things like:

- The variables in the Salary Data Set include Average full time adjusted salary per employee per year in a given country within the EU

- The values in the Data Set are in EUROS per year, it also includes the variable of the Time Period - 3 years in this case - 2022, 2023 and 2024. Another variable are the Countries within the EU
- Do the variables contain health information or SES information?
- Have they been measured by interviewing individuals or is the data coming from administrative sources?

Part 3 - Quantifying

3.1 Data cleaning

We perform several data cleaning steps to prepare the datasets for analysis: checking for missing values, removing incomplete observations, selecting relevant variables, and excluding aggregate regional entities.

```
colSums(is.na(gpg_data))
```

```
##      freq      unit  nace_r2      geo TIME_PERIOD      values
##         0         0         0         0         0         0
```

```
colSums(is.na(salary_data))
```

```
##      freq      unit      geo TIME_PERIOD      values
##         0         0         0         0         0
```

We checked both datasets for missing values. The output shows the number of missing values in each variable.

```
gpg_data <- na.omit(gpg_data)
salary_data <- na.omit(salary_data)
```

We used `na.omit()` to remove observations with missing values from both datasets.

```
gpg_data <- gpg_data %>%
  select(geo, TIME_PERIOD, values)
salary_data <- salary_data %>%
  filter(unit == "EUR") %>%
  select(geo, TIME_PERIOD, values)
```

We kept only the variables needed for the analysis (country, year, and values). For the salary dataset, only observations reported in euros (EUR) were retained.

```
gpg_data <- gpg_data %>%
  filter(!geo %in% c("EA19", "EA20", "EU27_2020"))

salary_data <- salary_data %>%
  filter(!geo %in% c("EA19", "EA20", "EU27_2020"))
```

Eurostat datasets often include aggregate entities such as “EA19” (Euro area with 19 countries), “EA20” (Euro area with 20 countries), and “EU27_2020” (European Union with 27 member states, 2020 composition). We exclude these to focus exclusively on individual member states, enabling country-level comparisons.

```
gpg_data <- gpg_data %>% rename(gpg = values)
salary_data <- salary_data %>% rename(salary = values)

merged_data <- left_join(gpg_data, salary_data, by = c("geo", "TIME_PERIOD"))
```

3.2 Generate necessary variables

Variable 1

```
northern_eu <- c("DK", "EE", "FI", "IE", "LT", "LV", "SE")
western_eu <- c("AT", "BE", "DE", "FR", "LU", "NL")
southern_eu <- c("CY", "ES", "GR", "HR", "IT", "MT", "PT", "SI")
eastern_eu <- c("BG", "CZ", "HU", "PL", "RO", "SK")

region_lookup <- c(
  setNames(rep("Northern Europe", length(northern_eu)), northern_eu),
  setNames(rep("Western Europe", length(western_eu)), western_eu),
  setNames(rep("Southern Europe", length(southern_eu)), southern_eu),
  setNames(rep("Eastern Europe", length(eastern_eu)), eastern_eu)
)

merged_data <- merged_data %>%
  mutate(region = region_lookup[geo])

merged_data <- merged_data %>%
  filter(!is.na(region))
```

```
merged_data <- merged_data %>%
  mutate(gpg_severity = (gpg / 100) * salary)

#cat("Merged data dimensions:", dim(merged_data), "\n")
#cat("Variables:", names(merged_data), "\n")
#head(merged_data)
```

Variable 2

```
region_summary <- merged_data %>%
  group_by(region, TIME_PERIOD) %>%
  summarise(
    mean_gpg = mean(gpg, na.rm = TRUE),
    mean_salary = mean(salary, na.rm = TRUE),
    mean_gpg_severity = mean(gpg_severity, na.rm = TRUE),
    n_countries = n(),
    .groups = "drop"
  )

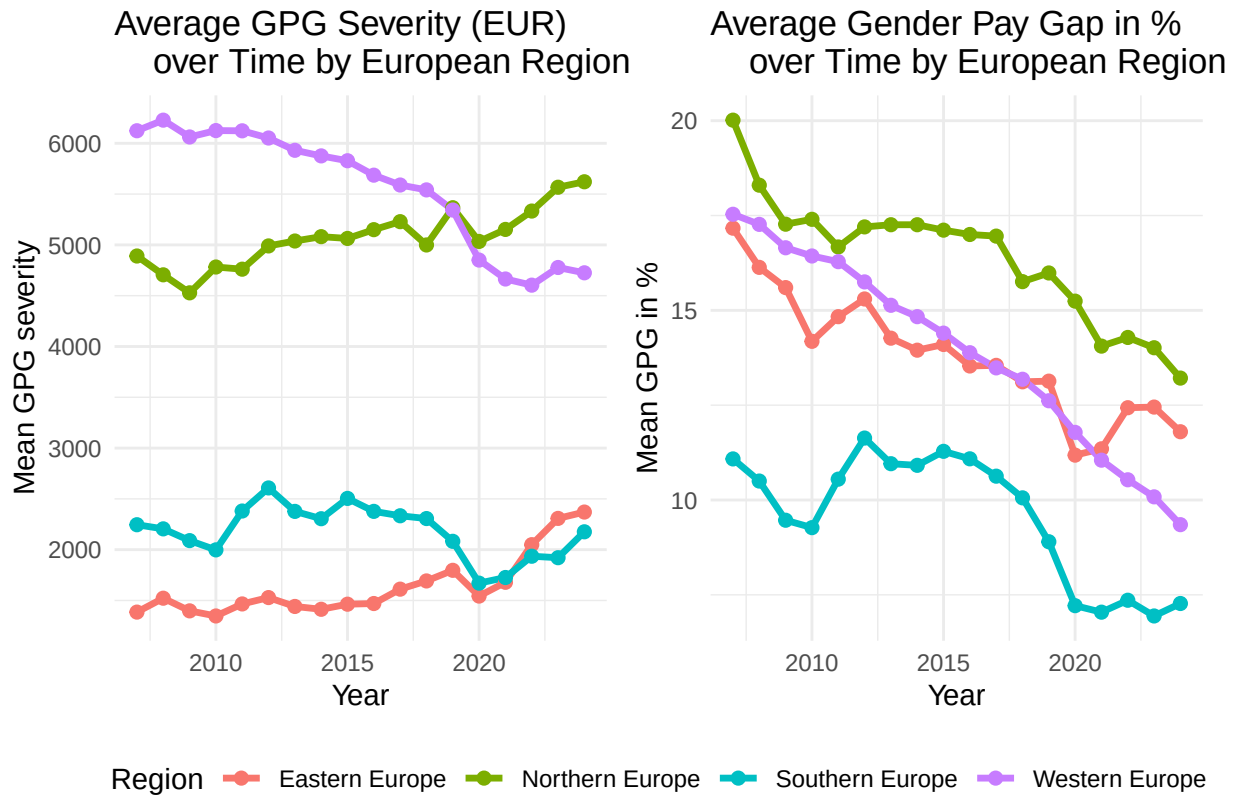
#cat("Region summary:\n")
#print(region_summary)
```

3.3 Visualize temporal variation

```
plot_severity <- ggplot(region_summary, aes(x = TIME_PERIOD, y = mean_gpg_severity, color = region, group = region)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Average GPG Severity (EUR)
    over Time by European Region",
    x = "Year", y = "Mean GPG severity", color = "Region"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")

plot_percentage <- ggplot(region_summary, aes(x = TIME_PERIOD, y = mean_gpg, color = region, group = region)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Average Gender Pay Gap in %
    over Time by European Region",
    x = "Year", y = "Mean GPG in %", color = "Region"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")

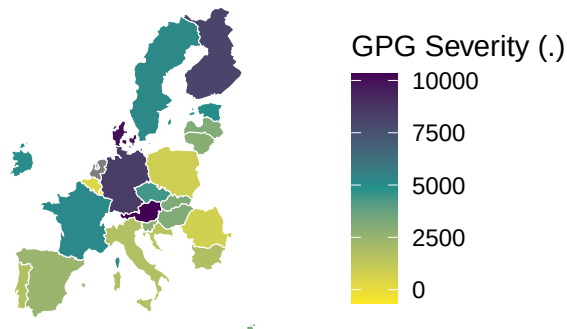
plot_severity + plot_percentage +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```



3.4 Visualize spatial variation

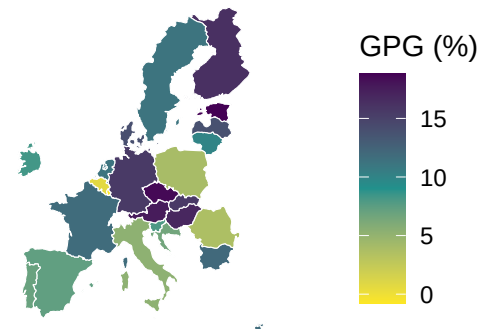
Gender Pay Gap Severity in 2024-01-01

Absolute Gap in Earnings (women vs men)



Gender Pay Gap Percentage in 2024-01-01

Percentage Difference in Average Earnings (women vs men)



Here you provide a description of why the plot above is relevant to your specific social problem.

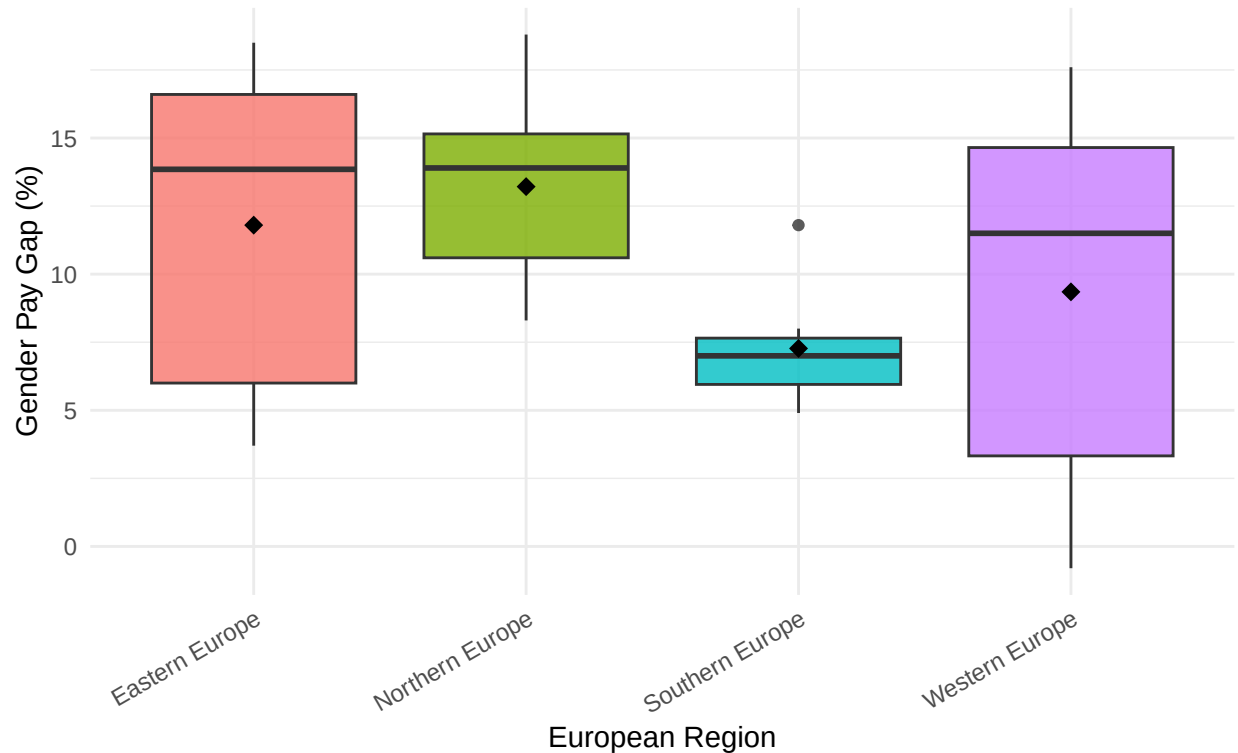
3.5 Visualize sub-population variation

What is the poverty rate by state?

```
gpg_latest %>%  
  ggplot(aes(x = region, y = gpg, fill = region)) +  
  geom_boxplot(alpha = 0.8) +  
  stat_summary(fun = mean, geom = "point", shape = 18, size = 3, color = "black") +  
  labs(  
    title = paste("Distribution of Gender Pay Gap by European Region (", latest_year, ")", sep = ""),  
    subtitle = "Diamond markers indicate regional means",  
    x = "European Region",  
    y = "Gender Pay Gap (%)",  
    fill = "Region"  
  ) +  
  theme_minimal() +  
  theme(  
    legend.position = "none",  
    axis.text.x = element_text(angle = 30, hjust = 1)  
  )
```


Distribution of Gender Pay Gap by European Region (2024-01-01)

Diamond markers indicate regional means



```
gpg_latest %>%
  group_by(region) %>%
  summarise(
    n = n(),
    mean_gpg = mean(gpg, na.rm = TRUE),
    median_gpg = median(gpg, na.rm = TRUE),
    sd_gpg = sd(gpg, na.rm = TRUE),
    min_gpg = min(gpg, na.rm = TRUE),
    max_gpg = max(gpg, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(desc(mean_gpg))
```

```
## # A tibble: 4 x 7
##   region          n mean_gpg median_gpg sd_gpg min_gpg max_gpg
##   <chr>      <int>   <dbl>     <dbl> <dbl>   <dbl>   <dbl>
## 1 Northern Europe     7    13.2      13.9  3.66     8.3    18.8
## 2 Eastern Europe     6    11.8      13.8  6.52     3.7    18.5
## 3 Western Europe     6     9.35      11.5  7.67    -0.8    17.6
## 4 Southern Europe     7     7.27       7    2.28     4.9    11.8
```

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

```
# COVID-19 event analysis  
# COVID-19 started in 2020, so we compare 2019 and 2021.  
# 2019 represents the pre-COVID period, while 2021 represents the post-COVID period.
```

```
gpg_covid_raw <- get_eurostat("tesem180", time_format = "num")
```

```
## indexed 0B in 0s, 0B/sindexed 1.00TB in 0s, 1.57PB/s
```

```
## Table tesem180 cached at /var/folders/zv/qz_pz5hn1wv9rx1z1nxp_n40000gn/T//RtmpuL83K5/eurostat/903f5
```

```
covid_event_data <- gpg_covid_raw %>%  
  filter(  
    TIME_PERIOD %in% c(2019, 2021),  
    !geo %in% c("EA19", "EA20", "EU27_2020"),  
    !is.na(values)  
  ) %>%  
  mutate(  
    region = region_lookup[geo],  
    covid_period = case_when(  
      TIME_PERIOD == 2019 ~ "Before COVID-19 (2019)",  
      TIME_PERIOD == 2021 ~ "After COVID-19 (2021)"  
    ),  
    covid_period = factor(  
      covid_period,  
      levels = c("Before COVID-19 (2019)", "After COVID-19 (2021)")  
    )  
  ) %>%  
  filter(!is.na(region))  
  
covid_event_summary <- covid_event_data %>%  
  group_by(region, covid_period) %>%  
  summarise(  
    mean_gpg = mean(values, na.rm = TRUE),  
    .groups = "drop"  
  )  
  
covid_event_summary
```

```
## # A tibble: 8 x 3  
##   region          covid_period      mean_gpg  
##   <chr>          <fct>          <dbl>  
## 1 Eastern Europe Before COVID-19 (2019)    13.1  
## 2 Eastern Europe After COVID-19 (2021)    11.4  
## 3 Northern Europe Before COVID-19 (2019)    16.0  
## 4 Northern Europe After COVID-19 (2021)    14.1  
## 5 Southern Europe Before COVID-19 (2019)     8.9  
## 6 Southern Europe After COVID-19 (2021)     7.04  
## 7 Western Europe Before COVID-19 (2019)    12.6  
## 8 Western Europe After COVID-19 (2021)    11.0
```

#For the event analysis, we use COVID-19 as the relevant event. COVID-19 began in 2020 and strongly affected European labour markets through lockdowns, remote work, reduced working hours, job losses, and increased unpaid care responsibilities. These changes may have affected men and women differently, making COVID-19 relevant for studying the Gender Pay Gap.

#This section uses two visualizations. The first figure shows the trend in the Gender Pay Gap around the COVID-19 period. The second figure focuses specifically on the comparison between 2019 and 2021, with 2019 representing the pre-COVID period and 2021 representing the post-COVID period.

```
gpg_covid_raw <- get_eurostat("tesem180", time_format = "num")
```

```
## Dataset query already saved in cache_list.json...
```

```
## Reading cache file /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//RtmpuL83K5/eurostat/903f5211762
```

```
## Table tesem180 read from cache file: /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//RtmpuL83K5
```

```
covid_trend_data <- gpg_covid_raw %>%
  filter(
    TIME_PERIOD >= 2017,
    TIME_PERIOD <= 2023,
    !geo %in% c("EA19", "EA20", "EU27_2020"),
    !is.na(values)
  ) %>%
  mutate(
    region = region_lookup[geo]
  ) %>%
  filter(!is.na(region))

covid_trend_summary <- covid_trend_data %>%
  group_by(region, TIME_PERIOD) %>%
  summarise(
    mean_gpg = mean(values, na.rm = TRUE),
    .groups = "drop"
  )

covid_comparison_summary <- covid_trend_summary %>%
  filter(TIME_PERIOD %in% c(2019, 2021))

covid_trend_summary
```

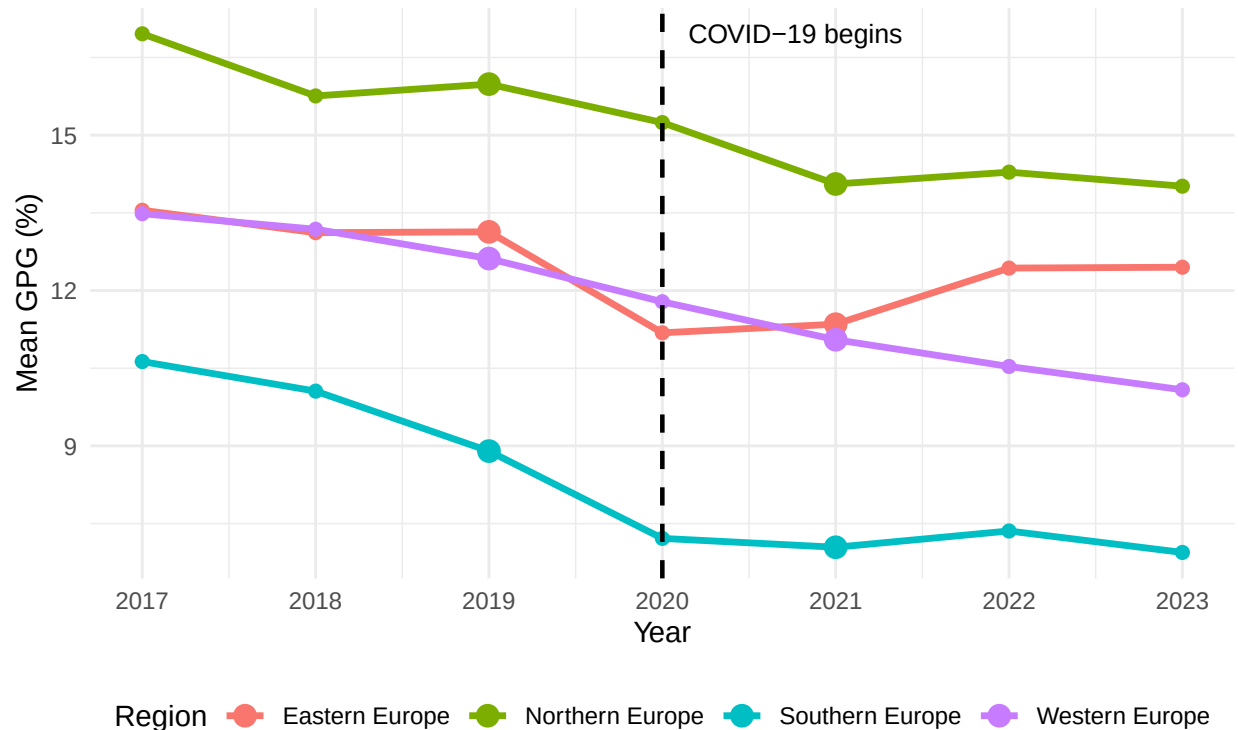
```
## # A tibble: 28 x 3
##   region      TIME_PERIOD mean_gpg
##   <chr>          <dbl>     <dbl>
## 1 Eastern Europe    2017      13.6
## 2 Eastern Europe    2018      13.1
## 3 Eastern Europe    2019      13.1
## 4 Eastern Europe    2020      11.2
## 5 Eastern Europe    2021      11.4
## 6 Eastern Europe    2022      12.4
## 7 Eastern Europe    2023      12.4
## 8 Northern Europe   2017      17.0
## 9 Northern Europe   2018      15.8
```

```
## 10 Northern Europe          2019      16.0
## # i 18 more rows
```

```
ggplot(covid_trend_summary, aes(x = TIME_PERIOD, y = mean_gpg, color = region, group = region)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  geom_point(
    data = covid_trend_summary %>% filter(TIME_PERIOD %in% c(2019, 2021)),
    size = 3.5
  ) +
  geom_vline(
    xintercept = 2020,
    linetype = "dashed",
    color = "black",
    linewidth = 0.8
  ) +
  annotate(
    "text",
    x = 2020.15,
    y = max(covid_trend_summary$mean_gpg, na.rm = TRUE),
    label = "COVID-19 begins",
    hjust = 0,
    size = 3.5
  ) +
  scale_x_continuous(
    breaks = 2017:2023
  ) +
  labs(
    title = "Gender Pay Gap Trend Before and After COVID-19",
    subtitle = "Average Gender Pay Gap (%) over time by European region, 2017-2023",
    x = "Year",
    y = "Mean GPG (%)",
    color = "Region"
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom"
  )
```

Gender Pay Gap Trend Before and After COVID-19

Average Gender Pay Gap (%) over time by European region, 2017–2023



```
gpg_covid_raw <- get_eurostat("tesem180", time_format = "num")
```

```
## Dataset query already saved in cache_list.json...
```

```
## Reading cache file /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//RtmpuL83K5/eurostat/903f5211762
```

```
## Table tesem180 read from cache file: /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//RtmpuL83K5
```

```
salary_raw <- get_eurostat("nama_10_fte", time_format = "num")
```

```
## indexed 0B in 0s, 0B/sindexed 1.00TB in 0s, 1.42PB/s
```

```
## Table nama_10_fte cached at /var/folders/zv/qz_pz5hn1wv9rzx1z1nxp_n40000gn/T//RtmpuL83K5/eurostat/2d
```

```
salary_clean <- salary_raw %>%
  filter(unit == "EUR") %>%
  select(geo, TIME_PERIOD, values) %>%
  rename(salary = values)

covid_data_severity <- gpg_covid_raw %>%
  filter(
    TIME_PERIOD >= 2017, TIME_PERIOD <= 2023,
    !geo %in% c("EA19", "EA20", "EU27_2020"),
```

```

    !is.na(values)
  ) %>%
  rename(gpg = values) %>%
  left_join(salary_clean, by = c("geo", "TIME_PERIOD")) %>%
  filter(!is.na(salary)) %>%
  mutate(gpg_severity = (gpg / 100) * salary)

covid_data_severity <- covid_data_severity %>%
  mutate(region = region_lookup[geo]) %>%
  filter(!is.na(region))

covid_trend_summary <- covid_data_severity %>%
  group_by(region, TIME_PERIOD) %>%
  summarise(
    mean_gpg_severity = mean(gpg_severity, na.rm = TRUE),
    .groups = "drop"
  )

names(covid_trend_summary)

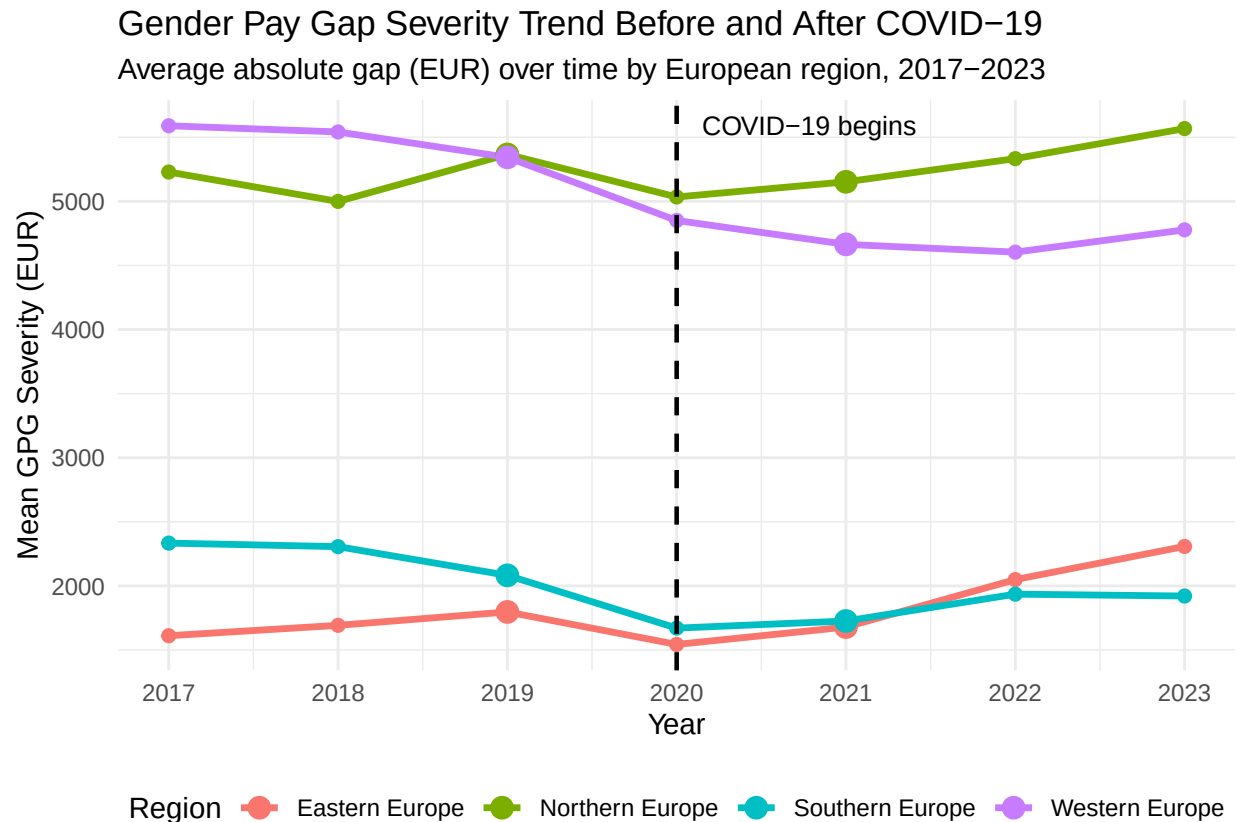
```

```
## [1] "region"          "TIME_PERIOD"      "mean_gpg_severity"
```

```

ggplot(covid_trend_summary, aes(x = TIME_PERIOD, y = mean_gpg_severity, color = region, group = region)) +
  geom_line(linewidth = 1.2) +
  geom_point(size = 2) +
  geom_point(
    data = covid_trend_summary %>% filter(TIME_PERIOD %in% c(2019, 2021)),
    size = 3.5
  ) +
  geom_vline(xintercept = 2020, linetype = "dashed", color = "black", linewidth = 0.8) +
  annotate(
    "text",
    x = 2020.15,
    y = max(covid_trend_summary$mean_gpg_severity, na.rm = TRUE),
    label = "COVID-19 begins",
    hjust = 0, size = 3.5
  ) +
  scale_x_continuous(breaks = 2017:2023) +
  labs(
    title = "Gender Pay Gap Severity Trend Before and After COVID-19",
    subtitle = "Average absolute gap (EUR) over time by European region, 2017-2023",
    x = "Year",
    y = "Mean GPG Severity (EUR)",
    color = "Region"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")

```



Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.

include=FALSE -> to be used when we want to hide code

```
renv::status()
```

```
## No issues found -- the project is in a consistent state.
```